

Shreyash Gadgil

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Education

Program	Institution/Board	%/CGPA	Year
M.S (Engineering Design: Robotics)	Indian Institute of Technology Madras Chennai, Tamil Nadu	8.56/10	2022-24
B.Tech. (Mechanical Engineering)	Sardar Patel College of Engineering Mumbai, Maharashtra	8.94/10	2015-19
Higher Secondary Leaving Certificate	Balaji Jr. college Ichalkaranji, Maharashtra	94.15%	2015
Secondary School Leaving Certificate	Vyankateshwara School Ichalkaranji, Maharashtra	94.73%	2013

Introduction

Robotics Systems Engineer with **5+ years** of experience designing, deploying, and integrating robotic systems across **manipulation, perception, autonomous navigation, and simulation-to-real** workflows. Deep expertise in **ROS/ROS2**, robotic manipulators (UR5e, SCARA), **RGB-D perception**, and **learning-based grasp planning**, with published research and hands-on production deployment experience. Proven ability to own full system integration mechanical, electrical, and software and bridge AI/ML models with reliable real-world robotic deployment in industrial and laboratory environments.

Professional Experience

1. **Principal Engineer — Robotics**

July 2024 – present

Technology Innovation Hub — IIT Bombay

- Automated Perovskite Deposition Platform Architected and deployed **DILBOT** (Devices and Interfaces Laboratory Robot) **India’s first fully automated perovskite thin-film deposition platform for lab** orchestrating a **Mitsubishi RH-6CH SCARA** (substrate handling) and a **Hamilton ZEUS X1 dual-pipette robot** (precision liquid dispensing) for end-to-end substrate processing from load to post-process XY scanning.
- Engineered **dual-robot synchronization** in Python with timed anti-solvent dispense callbacks (sub-second precision), coordinating parallel manipulation tasks: UV ozone treatment, dynamic/static spin coating, sequential annealing on dual hotplates, and automated XY batch scanning.
- Designed and fabricated custom grippers, mechanical end-effectors, and supporting hardware components; built a **configurable recipe execution framework with GUI** exposing 15+ per-recipe process variables (spin RPM, ramp/dwell times, dispense heights, trigger offsets, annealing durations) achieved **high repeatability** and increased researcher throughput by **over 5×**.
- Other Projects: developed **autonomous navigation** for a ground rover using a depth camera with lane-centering control; designed mechanical components for onboard data collection devices.
- Created **underwater vehicle simulation** in Gazebo integrated with MOOS-IvP, and extended the framework toward UAV deployment, enabling algorithm validation before physical deployment.
- Awarded **Employee of the Year 2024–25** for technical contribution and timely project deliverables and also Designed and delivered **ROS2 courses** (beginner and advanced) for graduate students, undergraduates, and industry professionals.

2. **Project Research Assistant**

July 2022 – July 2024

IIT M Pravartak Technology Innovation Hub

- Designed and deployed an **industrial automation solution for lithium-ion battery disassembly**, integrating **human-robot collaboration** to enhance efficiency and safety. **Owned full system integration** — hardware design, electronics, and software — from prototype to working deployment.
- Programmed a **UR5e** manipulator with **ROS** for camera-guided **pick-and-place**, combining real-time visual feedback with motion planning for reliable task execution in a semi-structured environment.

Worley India Pvt. Ltd.

- o Collaborated with cross-disciplinary teams to develop piping solutions and plant layouts for chemical and oil projects; contributed to engineering deliverables, 3D modelling, and cost-effective design.

Publications and Patents

1. *Journal Paper (Springer) and Patent (applied)* 2025

Grasp planning for under-actuated three-fingered grippers using a triangular representation

- o Proposed a **novel triangular grasp representation** enabling robust planning for **underactuated three-finger adaptive grippers**; validated on **RGB-D images** using a **modified ResNet50** deep learning model.
- o Validated end-to-end in **PyBullet simulation** and on real hardware using a **UR5e manipulator** with a **Robotiq 3-Finger Gripper**; demonstrated improved adaptability over conventional grasping strategies. **Patent filed.**

2. *IEEE Conference Paper Patent (applied)* 2024

Patent: Battery Disassembly System

Conference paper: Semi-Autonomous Robotic System for Efficient Recycling of Lithium-Ion Batteries

- o Developed a hybrid human-robot disassembly system for lithium-ion battery recycling, optimizing component recovery and reducing environmental waste. Received a **provisional patent**.

3. *Conference Paper* 2023

InnoGuideGPT: Integrating Conversational Interface and Command Interpretation for Navigation Robots

- o Integrated an LLM-based conversational interface with a navigation robot; validated in Gazebo simulation and deployed on **NVIDIA Jetson Nano** for cloud-based speech and LLM processing on a physical platform.

4. *IEEE Conference Paper* 2019

Investigating Application of Machine Learning for Recognition of Mechanical Engineering Drawings

- o Applied **Random Forest, KNN, SVM, and ANN** to recognize 2D machine component drawings (polygon identification); created datasets with controlled uncertainty for robust model evaluation.

Course Projects

Mobile Manipulator Motion Planning

Jan–May 2023

(M.S / ED5215 / Faculty: Prof. Dr. Bijo Sebastian)

IIT Madras

- o Explored **mobile manipulator path planning** using **RRT and RRT*** on a YOUTBOT in CoppeliaSim; developed a goal-to-goal controller and benchmarked planners on pick-and-place tasks.

Deep Learning Projects

Jan–May 2023

(M.S / CS6910 / Faculty: Prof. Dr. Mitesh Khapra)

IIT Madras

- o Built custom **feedforward/backprop** networks from scratch (92–95% MNIST accuracy); trained a **CNN from scratch** on iNaturalist (fine-tuned to 80%); developed a **PyTorch RNN** with attention for English-to-Marathi transliteration (80% character-level accuracy). Tracked experiments with **wandb.ai**.

Technical Skills

- o **Programming:** Python, C++
- o **AI / ML:** PyTorch, NumPy, Deep Learning, Classical ML
- o **Simulation:** Gazebo, CoppeliaSim, PyBullet
- o **Robot Hardware:** UR5e, Mitsubishi SCARA, KUKA iiwa 7, TurtleBot3
- o **System Integration:** Mechanical, electrical, software co-design
- o **Operating Systems:** Linux, Windows
- o **Frameworks:** ROS, ROS2, Git, GitHub
- o **Perception:** RGB-D cameras, depth sensors, pose estimation, grasping
- o **Sim-to-Real:** Model validation, hardware-in-the-loop, calibration
- o **Motion Planning & Control:** RRT, RRT*, PID, classical control, modern control, kinematics, dynamics
- o **CAD & Design:** SolidWorks, Fusion 360, Catia, Blender
- o **Domains:** Manipulation, AMR/navigation, industrial automation, lab robotics

Note: Highlighted text contains links to proofs and validation .